

Robustness Across Windows

M5 Retail Demand Forecasting — Problem Statement 11

Phase 9 — how big a difference has to be to matter

NPN AIA Hackathon — St. Joseph's College of Engineering

Phase 9 — the same models scored on four different 28-day periods. Generated 2026-08-14.

Terms. **RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.

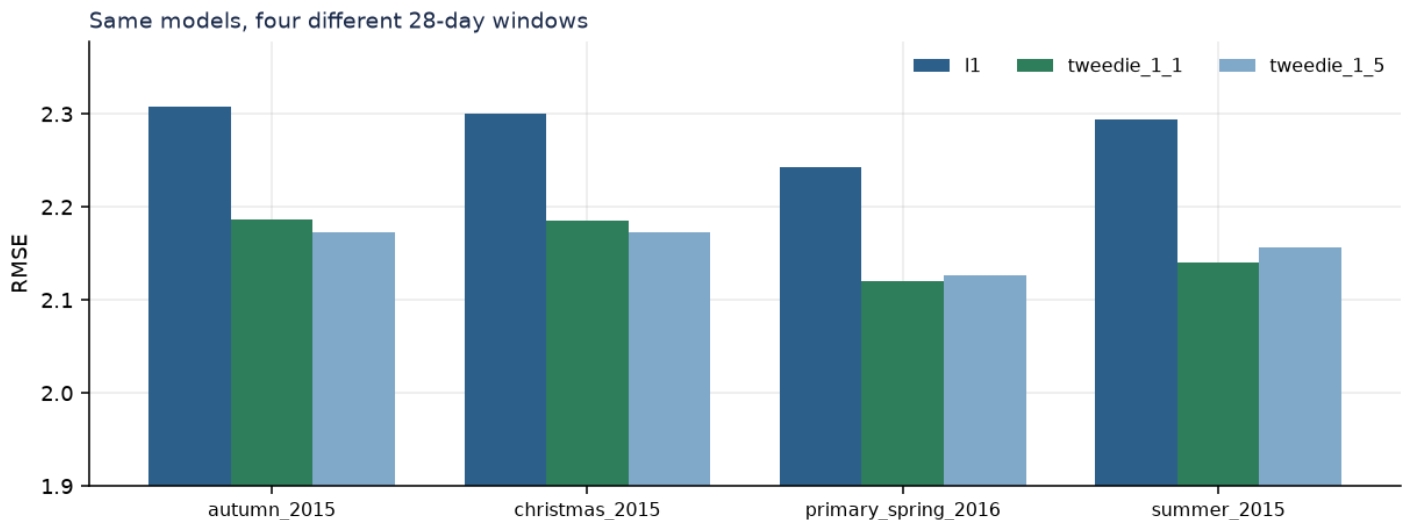
Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.

Why this is the most important phase

Several changes in this campaign looked like improvements on one window and reversed on another. This phase measures how large that window-to-window variation actually is — which tells us how big a difference has to be before it means anything.

Results (each model retrained per window)

Window	Dates	l1	tweedie_1_1	tweedie_1_5
autumn_2015	2015-10-02 .. 2015-10-29	2.3087	2.1869	2.1733
christmas_2015	2015-12-12 .. 2016-01-08	2.3005	2.1851	2.1731
primary_spring_2016	2016-04-25 .. 2016-05-22	2.2432	2.1210	2.1263
summer_2015	2015-07-16 .. 2015-08-12	2.2948	2.1405	2.1573



Consistency summary

Model	Mean RMSE	Std dev	Worst window	Mean MAE	MAE std
tweedie_1_5	2.1575	0.0221	2.1733	0.9767	0.0472
tweedie_1_1	2.1584	0.0329	2.1869	0.9818	0.0457
l1	2.2868	0.0296	2.3087	0.9068	0.0464

The finding that reframes the whole project

RMSE varies by roughly ± 0.033 across windows for the same model. Almost every "improvement" tested in this campaign was smaller than that.

Put plainly: the differences we have been chasing are smaller than the natural variation between one month and the next. That is why inner-window gains kept failing to transfer — they were noise, and our discipline of selecting on a separate window is what caught them.

It also puts the comparison with the team's benchmark in perspective. The disputed gap is 0.0886, only about 2.7x this natural window variation — and their validation window is unknown.